The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 104 - Safety Management Plan (SMP) (Page 48)

Regulation 104: Safety Management Plan.
(1) The owner, agent and manager of every mine shall, within one year of coming into force of these regulations, prepare and implement a comprehensive Safety Management Plan (SMP) for the mine based on Hazard Identification and Risk Assessment (HIRA).
(2) The Safety Management Plan shall include:
(a) A clear safety and health policy formulated by top management;
(b) Systematic identification of all hazards in mining operations, machinery, civil structures, and mine environment;
(c) Risk assessment of identified hazards using qualitative or semi-quantitative risk matrices (Likelihood vs Consequence);
(d) Specification of control measures conforming strictly to the hierarchy of controls: Elimination, Substitution, Engineering controls, Administrative controls, and Personal Protective Equipment (PPE);
(e) Allocation of duties and statutory responsibilities to qualified personnel (Mine Manager, Safety Officer, Overman, Mining Sirdar);
(f) Standard Operating Procedures (SOPs) for high-risk operations including blasting, slope stability, heavy machinery movement, and ventilation;
(g) Internal safety audits and monitoring protocols at predefined intervals;
(h) A comprehensive emergency response and disaster management plan with mock drill schedules;
(3) The Safety Management Plan shall be reviewed at least once every calendar year or immediately following any serious accident, dangerous occurrence, or major alteration in mining method or machinery.

The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 106 - Opencast Workings and Benches (Page 51)

Regulation 106: Manual and Mechanised Opencast Workings.
(1) In opencast workings in alluvium, soil, gravel, clay or other soft unconsolidated strata:
(a) The bench sides shall be sloped at an angle of safety not exceeding 45 degrees from the horizontal, or bench height shall not exceed 1.5 metres and width shall not be less than height;
(b) In mechanized operations with excavators in soft strata, bench height shall not exceed 3.0 metres unless specially permitted by the Chief Inspector.
(2) In hard rock opencast workings:
(a) The height of any bench shall not exceed the maximum reach or digging height of the machine in use for excavating;
(b) The width of any bench shall not be less than the height of the bench, and in mechanised opencast mines where heavy earth moving machinery (HEMM) like dumpers and shovel are deployed, the bench width shall not be less than the height of the bench plus 3 times the width of the widest machine operating on the bench;
(c) The overall pit slope angle shall be maintained in accordance with scientific geotechnical slope stability studies approved by CIM/DGMS;
(3) No person or equipment shall undercut any face or bench so as to cause overhanging or unstable strata.

The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 83 & 84 - Haul Roads and Dumpers (Page 41)

Regulation 83 & 84: Haul Roads, Dumper Movement, and Berm Standards.
(1) Design and maintenance of haul roads:
(a) The gradient of any haul road for dumper or truck operation shall generally not be steeper than 1 in 16. In exceptional circumstances for short ramps not exceeding 100 metres, gradient up to 1 in 10 may be permitted;
(b) The width of the haul road shall not be less than 3 times the width of the widest dumper or haul vehicle deployed in two-way traffic, plus an additional clearance of at least 1.5 metres;
(c) For one-way traffic, the width shall not be less than 2 times the width of the widest vehicle plus 1.0 metre;
(2) Parapet walls and Berms:
(a) Along the edge of every bench, haul road, spoil dump or ramp adjacent to a drop or drop-off exceeding 2 metres, strong earthen berms or crash barriers shall be maintained;
(b) The height of the berm shall not be less than the diameter of the tyre (wheel height) of the largest vehicle plying on the road (or at minimum 1.2 times tyre radius per DGMS Tech Circular 09/2008);
(c) The berm shall have a top width of not less than 1.0 metre and sides sloped at the natural angle of repose.

The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 153 & 155 - Standards of Ventilation and Mine Gases (Page 76)

Regulation 153 & 155: Standards of Ventilation and Monitoring of Inflammable and Noxious Gases.
(1) Standards of Ventilation:
(a) In every underground coal mine, the quantity of air in the general body of air shall not be less than 6.0 cubic metres per minute per person employed in the mine on the largest shift, or 2.5 cubic metres per minute per daily tonne of coal raised, whichever is larger;
(b) Air velocity at working faces shall be maintained between 0.5 m/s and 1.5 m/s to prevent stagnation and thermal stress.
(2) Gas limits in mine atmosphere:
(a) The percentage of oxygen in the ventilating current shall nowhere be less than 19.0%;
(b) The percentage of carbon dioxide (CO2) shall nowhere exceed 0.5%;
(c) In any coal seam of Degree I, II or III gassiness:
- Percentage of inflammable gas (methane / CH4) shall not exceed 0.75% in the general body of the return air of any ventilating district;
- Percentage of inflammable gas shall not exceed 1.25% in the main return airway;
(d) If inflammable gas concentration reaches 1.25% at any working place or blind heading, all electrical power must be cut off instantly and workers immediately withdrawn to a fresh air base.

The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 171 & 186 - Explosives, Shotfiring, and Blasting Precautions (Page 89)

Regulation 171 & 186: Shotfiring, Blasting Danger Zone, and Precautions.
(1) Opencast Blasting Danger Zone:
(a) Before firing any shot in an opencast mine, the shotfirer or blasting officer shall ensure that all persons within the Danger Zone?defined as a radius of not less than 500 metres from the blast site?have been evacuated to safe blasting shelters;
(b) All access roads and entrances leading into the 500-metre danger zone shall be effectively barricaded and manned by red-flag guards;
(2) Warning signals:
(a) Blasting siren protocol: First warning siren (continuous 2-minute siren) 10 minutes prior to blast; Second warning siren (intermittent 1-minute blasts) 2 minutes prior; All-clear siren (single 1-minute steady blast) after post-blast inspection confirms no misfires;
(3) Controlled Blasting:
(a) Where structures, villages, railways, or public highways fall within 500 metres, controlled blasting techniques (NONEL detonators, electronic delay detonators, air-decking, and heavy blasting wire mesh with conveyor-belt muffling) must be deployed;
(b) Ground vibration measured at nearest permanent structure shall not exceed DGMS circular vibration thresholds (e.g. 5 to 10 mm/s Peak Particle Velocity depending on dominant frequency).

The Coal Mines Regulations, 2017
Issuing Authority: Directorate General of Mines Safety (DGMS), Ministry of Labour and Employment, Govt of India
Document ID: DGMS-CMR-2017 | Status: In Force

Section: Regulation 149 - Precaution Against Inundation and Water Danger (Page 71)

Regulation 149: Precautions against danger from surface water and underground water-logging.
(1) No workings shall be extended to any point within 60 metres of any waterlogged workings, disused flooded mine, river, reservoir, tank or nallah, whether in the same mine or in adjoining mines, without prior written approval of the Chief Inspector of Mines.
(2) Where workings approach within 120 metres of known waterlogged or suspected inundated workings, advance exploratory boreholes shall be driven:
(a) At least one central advance borehole not less than 3 metres ahead of working face;
(b) Flank boreholes drilled at an outward angle of 30 degrees at regular intervals not exceeding 4.5 metres;
(3) Regular physical surveys and water level checks shall be conducted, and daily water discharge from mine sumps shall be logged.

DGMS (Tech) Circular No. 02 of 2020: Risk Assessment and SMP Formulation Guidelines
Issuing Authority: Directorate General of Mines Safety (DGMS), Dhanbad
Document ID: DGMS-CIRCULAR-2020-02 | Status: Active Advisory

Section: Section 3 - Systematic Hazard Identification and Risk Scoring (HIRA) (Page 2)

DGMS Tech Circular 02 of 2020: Guidelines on Risk Assessment Matrix for Mining Operations.
(1) Every mine operator must construct an objective 5x5 Risk Matrix combining Likelihood (1=Rare, 2=Unlikely, 3=Possible, 4=Likely, 5=Almost Certain) and Consequence Severity (1=Insignificant, 2=Minor injury, 3=Moderate/LTI, 4=Major/Permanent disability, 5=Catastrophic/Multiple fatalities).
(2) Risk Score = Likelihood x Consequence (Range 1 to 25).
- Low Risk (Score 1-4): Acceptable with routine monitoring and standard operating procedures;
- Medium Risk (Score 5-9): Tolerable; requires defined engineering controls and scheduled supervisor inspections;
- High Risk (Score 10-15): Unacceptable without additional risk reduction measures implemented within specified timeline;
- Critical/Extreme Risk (Score 16-25): Intolerable; work must cease immediately until controls reduce risk to Tolerable or Low.
(3) Risk Assessment Teams: Must comprise Mine Manager, Safety Officer, Geotechnical/Ventilation Engineer, Foremen/Overmen, Worker Representatives, and Trade Union Safety Committee delegates.

DGMS (Tech) Circular No. 02 of 2020: Risk Assessment and SMP Formulation Guidelines
Issuing Authority: Directorate General of Mines Safety (DGMS), Dhanbad
Document ID: DGMS-CIRCULAR-2020-02 | Status: Active Advisory

Section: Section 5 - Principal Mining Hazards Register (Page 4)

Principal Mining Hazards (PMH) mandated by DGMS:
Every Safety Management Plan must explicitly incorporate Principal Hazard Management Plans (PHMPs) for:
1. Ground Control and Slope Stability (Pit walls, benches, dump slope failures);
2. Inundation and Inrush of Water or Slurry;
3. Explosives, Blasting, Flyrock, and Ground Vibration;
4. Mobile Equipment, Haul Road Accidents, and Traffic Interactions;
5. Mine Fires, Spontaneous Combustion, and Coal Dust Explosion;
6. Toxic, Flammable, and Asphyxiating Gases and Underground Atmospheric Monitoring;
7. Electricity in Hazardous Environments and Flameproof Apparatus Integrity.